Rigidity of Cotangent Complex at Odd Primes

David Zhu

November 3, 2026

Contents

1 Introduction **2**

2 Outline **2**

3 Homotopy Operations on Simplicia $\mathbb{F}_p$-Algebras **4**

4 Detection in Quillen Spectral Sequence **5**

5 Nilpotence of Operations on Local Derived Fiber **6**

6 Chain Model **6**
6.1 Suspension on Chain Complexes 7

A Quillen Spectral sequence **8**

1 Introduction

Quillen stated his conjectures for essentially finite-type maps of Noetherian rings, using projective dimension; the flat-dimension formulation extends to arbitrary Noetherian homomorphisms. Under the finite-type hypotheses, the cotangent complex is pseudo-coherent, and finite flat dimension and finite projective dimension agree. The second conjecture adds the assumption $\mathrm{fd}_A B < \infty$ and predicts that $A \to B$ is locally complete intersection. Its conclusion is therefore the stronger bound one, under an additional hypothesis on the ring map itself.

This distinction is substantive. For a field k and $A = k[\epsilon]/(\epsilon^2)$, the augmentation $A \to k$ satisfies

$$L_{k/A} \simeq k[1] \oplus k[2], \qquad \operatorname{Tor}_i^A(k,k) \neq 0 \quad \text{for every } i \geq 0.$$

The cotangent calculation follows from transitivity for $k \to A \to k$, and the Tor calculation from the periodic resolution with differential given by multiplication by ϵ. Thus the bound two is sharp, and finite cotangent dimension does not imply finite flat dimension of the ring map. Moreover, the finiteness hypothesis concerns vanishing of $D_i(B/A; -)$ for all coefficient modules in sufficiently high degrees; boundedness of the ordinary homology $H_i(L_{B/A})$ alone is a different condition.

The complete-intersection conjecture was proved in characteristic zero by Avramov and Halperin, and in general by Avramov . Avramov's theorem applies to Noetherian homomorphisms locally of finite flat dimension. Turner obtained a different positive-characteristic proof using an algebraic analogue of Serre's theorem [**?**]. More recently, Briggs and Iyengar proved a stronger rigidity theorem: under the same local finite-flat-dimension hypothesis, vanishing of a single functor $D_n(B/A; -)$, for $n \geq 1$, already forces the map to be locally complete intersection [**?**, Theorem A]. These results do not remove the flat-dimension hypothesis. For the bound-two conjecture itself, earlier results include Avramov's theorem when either ring is locally complete intersection [**?**, Theorem 1.5], and the theorem of Avramov and Iyengar for algebra retracts, that is, surjections $A \to B$ admitting a ring section $B \to A$ [**?**, Theorem I].

Turner's work supplies the homotopical strategy used here [**?**, **?**, **?**]. One studies an augmented simplicial algebra over a residue field and compares two consequences of its origin and its André–Quillen homology. Its ring-theoretic origin imposes nilpotence on certain homotopy operations. A highest nonzero André–Quillen class in degree greater than two, on the other hand, can be detected by operations that remain nonzero under arbitrarily long iteration. Turner established the relevant characteristic-two detection results and proved the bound-two conjecture for Noetherian maps in characteristic two with Cohen–Macaulay source [**?**]. His detection argument uses relations among higher divided squares. As he explained in [**?**], the odd-primary relations available there did not give the admissible rewriting required by that proof. The issue was not the absence of odd-primary operations: Bousfield had already constructed them and computed the homotopy of free simplicial commutative algebras [**?**].

Two recent preprints also bear directly on this question. Schürg [**?**] proposes a proof in characteristic two without the Cohen–Macaulay assumption. Liu [**?**] claims the bound in positive residual characteristic, with statements extending beyond the equal-characteristic setting considered here. His argument also uses universal chain representatives and detection of operation towers in a free algebra [**?**, Lemma 3.7 and §4]. The present manuscript develops the nilpotence argument directly on the original local derived fiber using uniform syzygetic Artin–Rees, and treats detection for connected algebras with a finite-dimensional degree-one André–Quillen group. We cite these recent preprints for comparison; neither is an input to the proof below, and we make no claim of priority for the shared constructions.

2 Outline

Fix a morphism of Noetherian k-algebras $R \to A$. Standard localization techniques reduces the conjecture to the case where we consider a surjection of Noetherian local k-algebras $(R, \mathfrak{m}) \to (A, \mathfrak{n})$. In this case, the flat dimension of the cotangent complex can be detected by tensoring with the residue field

Proposition 2.0.1 (Stack 15.79.2)**.** Let R be a ring. Let $a, b \in \mathbb{Z}$. Let $K^\bullet$ be a pseudo-coherent complex of R-modules. The following are equivalent:

1. $K^\bullet$ is perfect with Tor amplitude in $[a, b]$.
2. For every maximal ideal $\mathfrak{m}$, we have $\mathrm{Tor}_i^R(K^\bullet; k(\mathfrak{m})) = 0$ for $i \notin [a, b]$

Thus, under perfectness assumption, it suffices to show $H_i^Q(A/R; k) = 0$ for $i > 2$. We will therefore suppress the coefficient k in Quillen homology from now on, unless other coefficients are used.

To prove the conjecture, Turner's program utilizes many tools from the homotopy theory of simplicial commutative algebras. For a k-vector space V, let $K(V, n)$ be the simplicial k-vector space characterized by having homotopy group concentrated in degree n, with value V.

Theorem 2.1. Given $A \in s\mathbf{CAlg}_k$, which is connected, there exists a sequence of cofibrations

$$A = A(0) \xrightarrow{j_0} A(1) \xrightarrow{j_1} A(2) \xrightarrow{j_2} ...$$

with the following properties

1. For each $n \geq 1$, $A(n)$ is $(n-1)$-connected.
2. For $s \geq n$,
$$H_s^Q(A) \cong H_s^Q(A(n))$$
3. There is a cofiber sequence
$$\mathrm{Sym}_k(K(H_n^Q(A), n)) \to A(n) \to A(n+1)$$

In particular, for A such that $H_i^Q(A)$ vanishes for $i > n$, we may identify

$$A(n) \cong \mathrm{Sym}_k(K(H_n^Q(A), n))$$

and obtain a comparison map

$$\phi_n : A \to \mathrm{Sym}_k(K(H_n^Q(A), n))$$

inducing isomorphism on H_n^Q. The work of Bousfield, Goerss and Lada identifies the homotopy groups of the right hand side, which is generated by compositions of homotopy operations.

Theorem 2.2 (Bousfield, 8.11)**.** Let $\{a\}_{a \in A}$ be $\mathbb{F}_p$-basis for $\pi_* X$. Then $\oplus_{s=0}^\infty \pi_t(\mathrm{Sym}^s X)$ is the non-unital associative graded-commutative $\mathbb{F}_p$-algebra with generators

$$\{\alpha_I(a) | a \in A, I \text{ admissible, } \mathrm{excess}(I) \leq |a|\}$$

In particular, one could find a composition of homotopy operations, of arbitrary length, acting non-trivially on the degree n-element in $\pi_*(\mathrm{Sym}_k(K(H_n^Q(A), n)))$. Now the proof reduces to two parts:

Theorem 2.3. Suppose $n > 2$. There is non-zero element in $y \in \pi_* A$ such that $(\phi_n)_*(y) = a$, and one could find a composition of homotopy operations, of arbitrary length, acting non-trivially on y.

The proof of the theorem is comparing the Quillen spectral sequences using ϕ_n.

Theorem 2.4. There is a convergent spectral sequence

$$E_{s,t}^1 := \pi_t(\mathrm{Sym}^s(H_*^Q(A/R; k))) \Rightarrow \pi_{s+t} X$$

This is the step where Turner had to restrict to characteristic 2: he needed explicit Adem relations, which is still absent to this day, among operations to control the differentials.

On the other hand, for $A = P \otimes_R k$, operations on $\pi_* A$ has the following nilpotence property

Theorem 2.5. There exists some integer N such that any composition of strictly suspended operation of length $> N$ on $\pi_* A$ is zero.

This is where Turner needed the assumption that A is Cohen-Macauley, so he could reduce to the Artinian case where the maximal ideal is nilpotent.

3 Homotopy Operations on Simplicia $\mathbb{F}_p$-Algebras

Theorem 3.1 (Bousfield, 8.11)**.** For an odd prime p, and a simplicial $\mathbb{F}_p$-algebra B, there exists two families of operations

$$v_t : \pi_n B \to \pi_{n+2t(p-1)} B \qquad 2 \leq 2t \leq n$$
$$\eta_t : \pi_n B \to \pi_{n+2t(p-1)+1} B \qquad 3 \leq 2t+1 \leq n$$

The operations have the following properties

1. The operations satisfy
$$v_t(a \cdot u) = a^p u$$
for $a \in \pi_0 B$
2. Except the top operation, all operations are additive. The top operation is the divided power.

Definition 3.1.1. For $k \geq 0$, let $I = (\delta_1, s_1, ..., \delta_k, s_k)$ be a $(2k)$-tuple of integers, where $\delta_i = 0, 1$ and $s_i \geq 1$. The index I is called **admissible** if $s_i \geq p s_{i+1} + \delta_i$ for $1 \leq i \leq k-1$. Its **excess** is defined to be

$$\text{excess}(I) := \begin{cases} 0 & k = 0 \\ 2s_1 + \delta_1 & k = 1 \\ 2s_1 + \delta_1 - \sum_{i=2}^{k} (2(p-1)s_i + \delta_i) & k \geq 2 \end{cases}$$

Define

$$\alpha_{\delta_i, s_i} := \begin{cases} v_{s_i} & \delta_i = 0 \\ \eta_{s_i} & \delta_i = 1 \end{cases}$$

and

$$\alpha_I := \alpha_{\delta_1, s_i} \circ ... \circ \alpha_{\delta_k, s_k}$$

If out of range just zero

Theorem 3.2 (Bousfield, 8.11)**.** Let $\{a\}_{a \in A}$ be $\mathbb{F}_p$-basis for $\pi_* X$. Then $\oplus_{s=0}^{\infty} \pi_t(\text{Sym}^s X)$ is the non-unital associative graded-commutative $\mathbb{F}_p$-algebra with generators

$$\{\alpha_I(a) | a \in A, I \text{ admissible}, \ \text{excess}(I) \leq |a|\}$$

In particular, the generators provide an infinite tower of non-zero operations.

Corollary 3.2.1. Let V be a k-vector space, and let u be the generating class in $\pi_n(\text{Sym}_k K(V, n))$. For any $j > 1$, the index $I_j := (0, p^j, 0, p^{j-1}, ..., 0, p, 0, 1)$ is admissible, and thus the operation $\theta_j := \alpha_{I_j}$ is non-zero on u for all j.

Lemma 3.3 (Degree-Weight Bound)**.** Let X be a simplicial k-module.

1. Suppose $d = \dim \pi_1(X) < \infty$. Then $\pi_t(\text{Sym}^s X)$ is zero if $t < 2s - d$.
2. Suppose $X = K(V, n)$. Then $\pi_t(\text{Sym}^s X)$ is zero if $t < 2s + n - 2$.

Proof. Let y be a non-zero monomial in $\pi_*(\mathrm{Sym}_k X)$. By Theorem 3.2, we can express y as product of k admissible monomial generators

$$y = \alpha_{I_1}(x_1)...\alpha_{I_k}(x_k)$$

where each I_j is admissible. Let $I_j := (\delta_1, s_1, ..., \delta_n, s_n)$. Admissiblity provides a lower bound on degree in terms of the weight

$$\begin{aligned}
\deg(\alpha_{I_i}(x_i)) &= \deg(x_i) + 2(p-1)\sum_{i=0}^{n-1} s_i \\
&\geq \deg(x_i) + 2(p-1)\sum_{i=1}^{n-1} p^i \\
&= \deg(x_i) + 2(p^n - 1) \\
&= \deg(x_i) + 2\mathrm{wt}(\alpha_I) - 2
\end{aligned}$$

Since degree and weight both add in the product, we get

$$\deg(y) \geq 2\mathrm{wt}(y) + \sum_{i=1}^{k} \deg(x_i) - 2$$

If $X = K(V, n)$, then $|x_i| = n$ for all i, and part (2) follows easily.

If $d = \dim(H_1^Q(Q)) < \infty$. We see only $|x_i| = 1$ will contribute to a negative summand, and each basis vector can only appear once in the product for it to be non-zero by graded commutativity. Therefore

$$\deg(y) > 2\mathrm{wt}(y) - d$$

so part (1) follows.

□

4 Detection in Quillen Spectral Sequence

Let A be an augmented simplicial $\mathbb{F}_p$-algebra. The Quillen spectral sequence reads

$$E^1_{s,t} := \pi_t(\mathrm{Sym}^s(H_*^Q(A))) \Rightarrow \pi_{s+t}X$$

Suppose n is the top non-vanishing degree of $H^Q(A)$, and let $u \in H_n^Q(A)$ be the generating class. Then, the element θ_j defined in Corollary 3.2.1 is an element in bidegree $(s, t) := (p^j, n + 2(p^j - 1))$. We can analyze the permanence property of this cycle:

Proposition 4.0.1. The element θ_j can be represented by a cycle $a \in F^s N(P)_t$, and $da \in F^{2s}N(P)_{t-1}$.

In particular, the spectral sequence differential d_r is trivial on the representing class a for $r < s$, as the target $F^{2s}N(P)_{t-1} \subseteq F^{s+r+1}N(P)_{t-1}$ is trivial in the subquotient $\frac{F^{s+r}N(P)_{t-1}}{F^{s+r+1}N(P)_{t-1}}$.

Proposition 4.0.2. The Quillen spectral sequence of $\mathrm{Sym}K(H_n^Q(A), n)$ degenerates on E^1.

For $r \geq s$, the target groups will be a subquotient of

$$E^1_{s+r,2s+n-3}$$

By Lemma 3.3, we see the target groups automatically vanish once we choose j large enough.

On the other hand, the there are no incoming differentials before page s, by comparing the spectral sequence with that of $\mathrm{Sym}K(H_n^Q(A), n)$. There are no incoming differentials on or after page s simply by degree reasons.

extension: by comparing the spectral sequences, we see that there is no extension problems: the difference lands in $\pi_{n+2s-2}\mathrm{Sym}^{s+1}(K(H_n))$. This group is zero by part (2) of Lemma 3.3.

5 Nilpotence of Operations on Local Derived Fiber

Let $(R, \mathfrak{m})$ be Noetherian local ring, M a finitely generated R-module. Choose a resolution of M be finite free R-modules

$$\ldots \xrightarrow{d_2} F_1 \xrightarrow{d_1} F_0 \xrightarrow{d_0} M \to 0$$

Theorem 5.1 (Uniform Artin–Rees for Syzygies)**.** For a fixed ideal $J \subseteq R$, there is a uniform integer $h \geq 0$ such that for every $n \geq h$, and $i \geq 0$, one has

$$J^n F_i \cap \mathrm{im}(d_{i+1}) \subseteq J^{n-h}\mathrm{im}(d_{i+1})$$

Take $J = \mathfrak{m}$, one has the following corollary

Corollary 5.1.1. There is a uniform integer $h \geq 0$, such that for every $q > h$ and $i > 0$, the reduction

$$\mathrm{Tor}_i^R(M; R/\mathfrak{m}^q) \to \mathrm{Tor}_i^R(M; k)$$

is zero.

Proof. Let $a \in F_i$ represent a cycle in $\mathrm{Tor}_i^R(M; R/\mathfrak{m}^q)$, i.e $da \in \mathfrak{m}^q F_i$. By Theorem 5.1, da is also contained in $\mathfrak{m}^{q-h}\mathrm{im}(d)$. We can thus find some $b \in \mathfrak{m}^{q-h}F_i$ with $d(a-b) = 0$. By exactness, there is some $a' \in F_{i+1}$ such that

$$da' = a - b$$

reducing modulo $\mathfrak{m}$, the class b vanishes, and thus a is a boundary mod $\mathfrak{m}$. □

Theorem 5.2 (Nilpotence on Local Derived Fiber)**.** There is an integer l such that all composition of additive operations of length greater than l vanishes on $\pi_*(P \otimes_R k)$.

Proof. Let $a \in N(P) \otimes_R k$ be a cycle, so that $da \in \mathfrak{m}N(P)$. Let $\alpha_I = v_{t_1}....v_{t_k}$ be a composition of a length l operation of the form v_t. By Lemma 6.2, $\alpha_I(a)$ is represented by iterated chain lifts $V_{t_1}...V_{t_k}(a) \in \mathfrak{m}^{p^l}N(P)$. By Corollary 5.1.1, taking $l > h$ will ensure the reduction mod $\mathfrak{m}$ is becomes a boundary, and therefore represents a trivial class in $\pi_*(P \otimes_R k)$.

□

6 Chain Model

Since the spectral sequence is defined via filtration on chains, we need to have a chain representative of the operations, which a priori is defined via universal homotopy classes, in order to analyze how the operations interact with the differentials. We begin with a quick review of the operations.

Suppose we are looking for homotopy operations on $\pi_* B$ for some simplicial k-algebra B

$$\pi_n B \to \pi_{n+t} B$$

By representability and adjunction, these operations are represented by classes in $\pi_{n+t}(\mathrm{Sym}K(k,n))$. Let $a : K(k,n) \to B$ be a representative of some class $\tilde{a} \in \pi_n B$, and $u : K(k, n+t) \to \mathrm{Sym}(K(k,n))$ to be a representative of $\tilde{u} \in \pi_n B$. The evaluation $u(a) \in \pi_{n+t}B$ is given by the composition

$$K(k, n+t) \xrightarrow{u} \mathrm{Sym}(K(k,n)) \xrightarrow{a} B$$

Under the Dold-Kan correspondence, operations can be understood more explicitly: they are classes in $H_{n+t}(N(\mathrm{Sym}K(k,n)))$, which can be represented by some chain map

$$k[n+t] \to N(\mathrm{Sym}K(k,n))$$

which corresponds to an explicit degree $n+t$ cycle $z_{n+t} \in N(\mathrm{Sym}K(k,n))$.

6.1 Suspension on Chain Complexes

Let C be the chain complex concentrated in degree i. There is an exact sequence of chain complexes

$$0 \to C \to D \to C[1] \to 0$$

where $D := \mathrm{Cone}(\mathrm{id}_C)$ is the cone, which is acyclic. Then the connecting morphism

$$\partial : H_j(C[1]) \to H_{j-1}(C)$$

is the suspension isomorphism. In the case $C = k[n-1]$ and under Dold-Kan, the exact sequence becomes

$$0 \to K(k,n-1) \to K(D_n) \to K(k,n) \to 0$$

We shall refer to these as **additive suspension**. The non-additive case happens when we apply the reduced symmetric algebra functor: we have the commutative diagram of simplicial commutative non-unital algebras

$$\begin{array}{ccccc} \mathrm{Sym}K(k,n-1) & \xrightarrow{i_*} & \mathrm{Sym}K(D_n) & \longrightarrow & \mathrm{Coker} \\ & & \downarrow q & \nearrow & \\ & & \mathrm{Sym}K(k,n) & & \end{array}$$

Note that $\mathrm{Sym}K(D_n)$ is still acyclic. After normalization, a class $z \in N(\mathrm{Sym}K(k,n-1))$ still injects to a class $i_*(z)$, which is a boundary. Choose ω such that $d\omega = i_*(z)$, and define

$$\sigma z = q(w) \in N(\mathrm{Sym}K(k,n+1))$$

It is clear that $q(w)$ is a cycle since w is coming from the kernel, and the choice of ω does not affect the homology class of σz.

Theorem 6.1 (Bousfield)**.** Suppose z_{pn} is the normalized cycle representing the top divided power operation, coming from the universal generator in

$$v_n \in \pi_{pn}(\mathrm{Sym}^p K(k,n))$$

Then, σz_{pn} represents the additive operation from the generator

$$\pi_{pn+1}(\mathrm{Sym}^p K(k,n+1))$$

The input degree and output degree are both raised by 1, and the resulting operations is no longer the top operation. In other words, all the additive operations in v_t family are all suspended from the top divided powers from lower input degree.

Lemma 6.2 (Chain Lift with Controlled Differential)**.** Let B be a simplicial commutative $\mathbb{F}_p$ algebra. Let $a \in N(B)_n$ and $2j < n$. There exists a chain lift

$$V_j^n(a) \in N(P)_{n+2j(p-1)}$$

satisfying

1. If a is a cycle, then $V_j^n(a)$ represents $v_j(a)$.
2. If P is an algebra over discrete commutative ring R, $J \subseteq R$ an ideal, and $da \in J^m(N(B))_{n-1}$

$$dV_j^n(a) \in J^{pm}(N(B))_{(n-1)+2j(p-1)}$$

3. If P is augmented over k, with augmentation ideal I, and $a \in N(I^s)_n$ with $da \in N(I^t)_{n-1}$ for some $t \geq s$, then

$$V_j^n(a) \in N(I^{ps})_{n+2j(p-1)}, \qquad dV_j^n(a) \in N(I^{pt})_{(n-1)+2j(p-1)}$$

Proof. Let v be the generator in n, and u be the generator in degree $n-1$ of D_n, with $dv = u$. There is a simplicial k-module map

$$D_n \to N(B)$$

by $v \mapsto a$, $u \mapsto da$. Since $da = 0$, we see the induced normalized algebra map a_* factors through q

$$\begin{array}{ccc} N(\mathrm{Sym}^p K(k,n-1)) & & \\ \downarrow{\scriptstyle i_*} & \searrow^{f_u} & \\ N(\mathrm{Sym}^p K(D_n)) & \xrightarrow{f} & N(B) \\ \downarrow{\scriptstyle q} & \nearrow_{f_v} & \\ N(\mathrm{Sym}^p K(k,n)) & & \end{array}$$

Let $z_j^n \in N(\mathrm{Sym}K(k,n))_{n+2j(p-1)}$ be the cycle representing the operation v_j. By Theorem 6.1, z_j^n is suspended from some $z_j^{n-1} \in N(\mathrm{Sym}K(k,n-1))_{(n-1)+2j(p-1)}$. Choose ω_j^n such that $d(\omega_j^n) = z_j^{n-1}$, with $q(\omega_j^n) = z_j^n$, we define

$$V_j^n(a) := f(\omega_j^n)$$

It is immediate from construction that $V_j^n(a)$ represents the class $v_j(a)$. For (2), we have by construction

$$dV_j^n(a) = f(d\omega_j^n) = f_u(z_j^{n-1})$$

The map f_u is induced by sending the degree $n-1$ generator u to $da \in J^m((N(B)))_{n-1}$; the generator z_j^{n-1} is of the form $s_1(u) \otimes ... \otimes s_p(u)$, where $s_i(u)$ is some linear combination of the image of u under iterated degeneracy maps. The image $d_u(z_j^n)$ is then

$$d_u(z_j^n) = s_1(da)...s_p(d_a)$$

since the degeneracy maps are R-linear, and each $da \in J^m B$, it follows that $d_u(z_j^n)$ is in $J^{pm} \in B$. It is clear that normalization respects the ideal filtration. (3) follows from exactly the same argument.

□

A Quillen Spectral sequence

By the standard model structure on $s\mathbf{CAlg}_R$, one may choose a cofibrant replacement $P \to A$ such that P is semi-free, i.e every P_n is a free R-algebra. Let $X := P \otimes_R k$. Denote by I the augmentation ideal and $QX := I/I^2$ the indecomposables. Standard computation gives

$$QX \cong \Omega_{P/R} \otimes_P k$$

And therefore we have $H_i^Q(A/R) \cong \pi_i(QX)$. This provides the interpretation that Quillen homology of an augmented simplicial k-algebra X is the homotopy group of its indecomposables QX. we have a descending filtration of X by

$$X \supseteq I \supseteq I^2 \supseteq I^3 \supseteq ...$$

which leads to a spectral sequence

$$E^1_{s,t} := \pi_t(I^s/I^{s+1}) \Rightarrow \pi_{s+t}X$$

A standard fact on the model structure on $s\mathbf{CAlg}_R$ says one may choose P to be semi-free, meaning each simplicial degree P_n is a free R-algebra. It follows that

$$I^s/I^{s+1} \cong \mathrm{Sym}^s_k(I/I^2)$$

Since a chain complex of vector spaces splits as direct sums of its homology, I/I^2 also splits as the direct sum of its homotopy groups under the Dold-Kan correspondence.